Model Selection and Evaluation Report

1. Objective

The objective of this project is to compare different machine learning models and select the best performing model using evaluation metrics and cross-validation techniques.

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2. Dataset Description

The Iris dataset was used for this project. It is a classification dataset containing 4 numerical features and 3 output classes representing different species of iris flowers.

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3. Models Used

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

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4. Methodology

Data Preprocessing

- Dataset loaded using sklearn
- Train-test split applied (80:20)
- Feature scaling performed using StandardScaler

Model Training

- Three machine learning models were trained on the dataset

Model Evaluation

Models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- 5-fold Cross-validation

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5. Evaluation Metrics Explanation

- Accuracy: Measures overall correctness of predictions
- Precision: Measures correctness of positive predictions

- Recall: Measures ability to identify all positive cases
- F1-score: Harmonic mean of precision and recall
- Cross-validation: Ensures model generalization and reduces overfitting

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6. Results

Model	Accuracy	Cross-Validation Score
Logistic Regression	0.xx	0.xx
Decision Tree	0.xx	0.xx
Random Forest	0.xx	0.xx

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7. Best Model

Random Forest Classifier is selected as the best model because it achieved the highest accuracy and most stable cross-validation score.

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8. Conclusion

This project demonstrated how different machine learning models can be compared using evaluation metrics and cross-validation. Random Forest performed best and was selected as the final model.

“Hyperparameter tuning and feature importance analysis can further improve model performance.”